

Program : Diploma in Computer Application and Business Management	
Course Code : 3254	Course Title: Quantitative Techniques
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives

- To provide a fundamental knowledge in statistics.
- To be able to develop the measures of central tendency and dispersion, and solve related problems, apply the usage of correlation, regression and Index numbers to real life situations.

Course Prerequisites:

Topic	Course code	Course name	Semester
Probability		Business Mathematics	2

Course Outcomes:

On completion of the course, students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basics of statistics and summarize the concept of statistical investigation	14	Understanding
CO2	Apply the measures of central tendency for problem solving	17	Applying
CO3	Use the concepts of correlation and regression for business decision making	14	Applying
CO4	Utilize index numbers to solve business problems	13	Applying
	Series Exam	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basics of statistics and summarize the concept of statistical investigation.		
M1.01	Define statistics & list the objectives of statistics.	2	Understanding
M1.02	Outline the functions & limitations of statistics.	1	Understanding
M1.03	Recall the meaning of Statistical Investigation & explore steps involved in statistical investigation.	3	Understanding
M1.04	Explain the methods of collecting Primary data & Secondary data	4	Understanding
M1.05	Explain the meaning and importance of classification and tabulation	4	Understanding
Contents: Meaning of statistics - definition - objectives - functions - importance - limitation - statistical investigation - meaning and definition - preliminaries to the collection of data - types of data - primary data - secondary data - classification and tabulation - meaning and definition - objectives of classification - kinds of classification - requisites of good classification - tabulation - meaning and definition of tabulation - essentials of good tabulation			
CO2	Apply the measures of central tendency for problem solving.		
M2.01	State the meaning and importance of various measures of central value	2	Understanding
M2.02	Define Mean, Median & Mode	1	Understanding
M2.03	Enumerate the requisites of good average, advantages and disadvantages of Mean, Median, Mode	2	Understanding

M2.04	Solve problems on Arithmetic Mean, Median, Mode.	5	Applying
M2.05	Describe weighted arithmetic mean and solve problems	2	Applying
M2.06	Explain the measures of dispersion	1	Understanding
M2.07	Solve problems on range, standard deviation and variance	4	Applying
	Series Test - I	1	
Contents: Measures of central tendency - Mean, Median, Mode - requisites of good average - advantages & disadvantages - Solve problems on Arithmetic Mean, Median, Mode - Weighted arithmetic mean - Explain the measures of dispersion, Solve problems on range, standard deviation and variance			
CO3	Use the concepts of correlation and regression for business decision making.		
M3.01	Understand the meaning and importance of correlation	2	Understanding
M3.02	Classify the types of Correlation	2	Understanding
M3.03	Explain the measurement of correlation-simple and rank correlation	1	Understanding
M3.04	Solve problems on simple and rank correlation	6	Applying
M3.05	Explain Linear Regression and solve problems	3	Applying
Contents: Correlation - importance of correlation - types of correlation - measurement of correlation - simple and rank correlation - Solve problems on simple and rank correlation - Linear Regression - Solve problems			
CO4	Utilize index numbers to solve business problems.		
M4.01	Understand the concept of Index Numbers & enumerate the uses of Index Numbers.	2	Understanding
M4.02	Summarize the uses of index numbers and the problems in construction of index numbers	2	Understanding
M4.03	Understand various methods of constructing index numbers	1	Understanding

M4.04	Solve problems on Index Numbers.	8	Applying
	Series Test - II	1	

Contents:

Index Numbers - meaning and definition - Uses of index numbers - Problems in construction of index numbers - methods of constructing index number - Solve problems on Index Numbers.

Text / Reference:

T/R	Book Title/Author
R1	S.P.Gupta, Fundamentals of Statistics , S.Chand & Co, Delhi
R2	L.R. Potti, A text book of Business Statistics , Thomson publication s
R3	B.L. Agarwal, Basic Statistics , New Age International Publishers, Sixth Edition
R4	Dr. S. M. Shukla & Dr. Hina Agarwal, Fundamentals of Statistics , Sahitya Bhavan Publications

Online Resources:

Sl.No	Website Link
1	http://www.textbooksfree.org/Statistics%20Book.htm
2	https://lecturenotes.in/subject/214/quantitative-techniques-qt
3	http://www.universityofcalicut.info/syl/QuantitativeTechniquesforBusiness.pdf
4	https://www.tutorialsduniya.com/notes/quantitative-techniques-for-management-notes/